

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 4: L5

Question 1 (Lecture 5)

In a linear programming (LP) problem, which of the following must always be true?

- (a) The feasible region is bounded.
- (b) The objective function and all constraints are linear.
- (c) The optimal solution is always at the intersection of all constraints.
- (d) The constraints can be strict inequalities as well as non-strict inequalities.

Ans: (b)

Question 2 (Lecture 5)

True or false: If a linear program (LP) is neither infeasible nor unbounded, it is finitely optimal.

- (a) True
- (b) False

Ans: (a)

Question 3 (Lecture 5)

A company wants to maximize its monthly profit by deciding how many units of two products to produce each month.

Products: Product A and Product B

Profit per unit: \$40 for Product A, \$30 for Product B

Constraints:

- Product A requires 2 hours of machine time per unit.
- Product B requires 1 hour of machine time per unit.
- The company has at most 100 hours of machine time available each month.
- To fulfill a contract, the company must produce at least 10 units of Product B each month.
- The company has limited storage space and shipping capacity, allowing it to handle only 40 finished units per month.

- (i) Formulate the problem mathematically
- (ii) Solve using graphical method (show feasible region and optimal point)
- (iii) Implement in Python using CVXPY or PuLP library

Ans: (i)

Maximize $z = 40x_A + 30x_B$ (total profit)

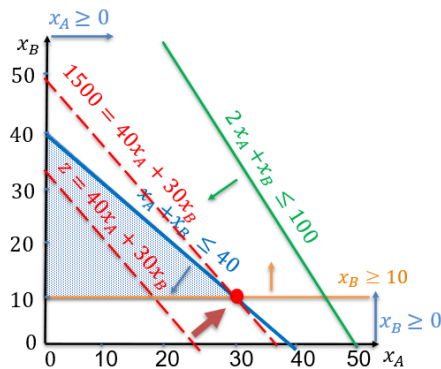
Subject to: $2x_A + x_B \leq 100$ (machine time limit)

$x_B \geq 10$ (minimum requirement for Product B)

$x_A + x_B \leq 40$ (production capacity)

$x_A, x_B \geq 0$ (non-negativity)

(ii)



Evaluate three vertices: (0, 10), (0, 40), (30, 10):

vertex (x_A, x_B)	$z = 40x_A + 30x_B$
(0, 10)	300
(0, 40)	1200
(30, 10)	1500

After evaluating all vertices, the maximum profit: 1500 ($x_A = 30, x_B = 10$)

Note: The solution is based on the fundamental theorem in LP: In linear programming, if an optimal solution exists, it occurs at a vertex (or corner point) of the feasible region.

(iii) **Ans:** Please refer to TUT4-Q3.py for sample implementation. The maximum profit: 1500 ($x_A = 30, x_B = 10$)

Question 4 (Lecture 5)

Given the linear program:

Minimize $z = 2x_1 + x_2$

subject to:

$$x_1 + x_2 \geq 5$$

$$x_1 + 2x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

The program is:

(a) Bounded and feasible

- (b) Feasible but not bounded
- (c) Bounded but not feasible
- (d) Neither bounded nor feasible

Ans: (a)

Question 5 (Lecture 5)

Resource Allocation Problem

Imagine you are a Year 3 student planning your semester courses. Your goal is to maximize your benefit by carefully selecting the best combination of courses, but you have several important limits to consider:

- You can take up to 20 credits in total for the semester.
- Your weekly workload should not exceed 40 hours to maintain a healthy balance.
- You need to avoid time slot conflicts so your classes don't overlap.
- Due to personal and cognitive limits, you want to take no more than 3 courses.
- You can only enroll in whole courses—no partial enrollment allowed.

The course information is shown in the table below.

How would you select your courses to get the most academic benefit while respecting these constraints? Describe your approach to solving this problem. You may use any method (e.g., brute-force, tools like Python). Justify your solution.

ID	Course	Credits	Workload per week	Time Slot	Benefit score*
1	EE2213	4	10 hrs	Mon 10–12	9
2	MA2101	4	9 hrs	Mon 10–12	7
3	CDE2212	4	11 hrs	Tue 2–5	8
4	CE3201	2	5 hrs	Wed 3–5	5
5	EE3801	3	7 hrs	Thu 1–3	6

* Suppose the benefit score for each course is estimated based on a combination of factors such as academic return, personal interest, and career relevance.

Ans: Formulating the problem:

Decision variables:

$x_i = 1$ if course i is selected, 0 otherwise, $i = 1, \dots, 5$

Objective Function (maximize academic benefit):

Maximize $Z = 9x_1 + 7x_2 + 8x_3 + 5x_4 + 6x_5$

subject to:

- *Total credits:* $4x_1 + 4x_2 + 4x_3 + 2x_4 + 3x_5 \leq 20$
- *Total weekly workload:* $10x_1 + 9x_2 + 11x_3 + 5x_4 + 7x_5 \leq 40$
- *No time conflicts:* $x_1 + x_2 \leq 1$ (EE2213 and MA2102 conflict (both Mon 10-12))
- *Total no. of courses:* $x_1 + x_2 + x_3 + x_4 + x_5 \leq 3$
- *Binary decision:* $x_i \in \{0, 1\} \quad i = 1, \dots, 5$

Please refer to TUT4-Q5.py for sample implementation.

Ans: Maximum total benefit score: 23 (select CDE2212, EE2213, EE3801)